

Section-A (40 Marks)

1. What are the principal constituents of cement? Briefly describe. [2+3=5]

Answer:

Cement is made up of various chemical constituents in definite proportions, which control the strength, setting, and durability of cement. The principal constituents of the cement are listed and described in brief:

i. *Lime* (CaO):

The most important constituent of cement is lime which is about 60-65%

It provides strength and soundness to cement. However, excess lime causes the cement to expand and crack.

ii. *Silica* (SiO_2):

Cement consists about 17-25% of silica.

It imparts the strength and durability of cement by forming silicates such as dicalcium silicates.

Excess silica causes slow setting and hardening.

iii. *Alumina* (Al_2O_3):

The content of alumina in the cement is about 3-8%.

It helps in quick setting of cement and reduces clinkering temperature.

Excess alumina in cement results in reduced resistance to sulphate attack.

iv. *Iron Oxide* (Fe_2O_3):

Cement consists of 0.5-6% of iron oxide.

It gives color to the cement and contributes in its strength and hardness.

v. *Magnesia* (MgO):

Cement consists of 0.1-4% of magnesia.

Magnesia imparts hardness to the cement, however excess magnesia results in unsoundness of cement.

vi. *Gypsum* ($CaSO_4$):

The content of gypsum in the cement is about 2-5%.

Gypsum helps to control the setting time of the cement and prevents its flash setting.

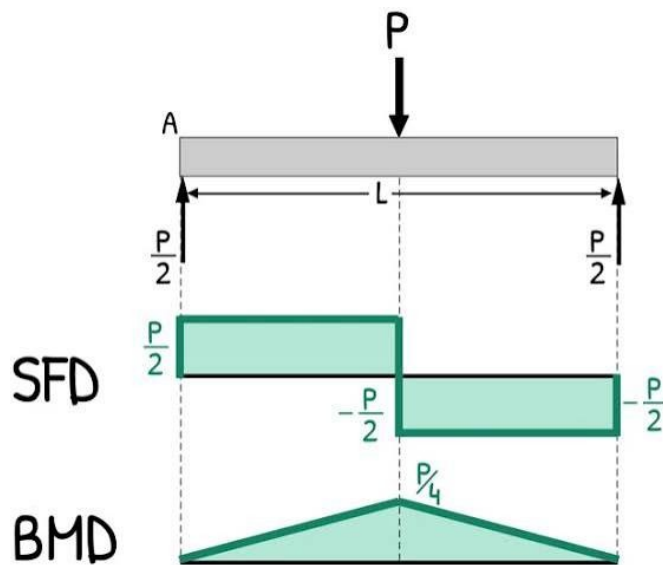
Thus, the constituents are to be mixed in required proportion for the manufacture of cement which ensures desired strength, setting behavior, and durability of cement.

2. Define the relationship between loads, shear and bending moment. [2+3=5]

Answer:

Load, shear force, and bending moment are the fundamental parameters used to analyze the behavior of beams under various loading conditions.

1. **Load (w):** Load is the external force acting on a beam. It can be a point load (kN) or distributed load (kN/m).
2. **Shear force (V):** Shear force at a section is the algebraic sum of all vertical forces acting on either side of that section.
3. **Bending moment (M):** Bending moment at a section is the algebraic sum of moments of all forces about that section.



The relationship among load, shear force and bending moment as:

a. Relationship between load and shear force:

The rate of change of shear force with respect to the beam length is equal to the negative of the distributed load.

$$\frac{dV}{dx} = -w$$

or

$$w = -\frac{dV}{dx}$$

where :

$w = \text{distributed load intensity (kN / m)}$

$V = \text{shear force (kN)}$

$x = \text{distance along the beam (m)}$

If there is no distributed load ($w=0$), the shear force remains constant.

A uniformly distributed load (UDL) causes the shear force diagram to be a straight line.

A point load causes a sudden jump in the shear force diagram.

b. Relationship between shear force and bending moment

The rate of change of bending moment with respect to the beam length is equal to the shear force.

$$\frac{dM}{dx} = V$$

or

$$M = \int V dx$$

where :

$M = \text{bending moment (kN}\cdot\text{m)}$

$V = \text{shear force (kN)}$

If the shear force is zero, the bending moment is maximum or minimum.

A constant shear force produces a straight-line bending moment diagram.

A linearly varying shear force produces a parabolic bending moment diagram.

Combined Relationship:

$$w = -\frac{dV}{dx}$$

$$V = \frac{dM}{dx}$$

Or, equivalently

$$w = -\frac{d^2M}{dx^2}$$

Thus, load, shear force, and bending moment are interrelated quantities, and their relationships are essential for structural analysis and safe design of beams.

3. Classify the loads acting on the structures and explain any two. [2+3=5]

Answer:

Structural loads are those forces that act on a structure during its construction and service life. According to common structural design practice, the loads are classified as follows:

- i. Dead Load (DL)
- ii. Live or Imposed Load (LL)
- iii. Wind Load (WL)
- iv. Earthquake (Seismic) Load (EL)
- v. Snow Load
- vi. Temperature Load
- vii. Impact or Dynamic Load
- viii. Earth and Water Pressure
- ix. Settlement or Deformation Load

Dead load is the permanent load due to the self-weight of the structure and all fixed components attached to it.

Examples:

- a. Weight of beams, columns, slabs, and walls
- b. Roof covering
- c. Flooring and plaster
- d. Fixed equipment and permanent partitions

Characteristics:

- a. Dead load acts continuously on the structure throughout its service life.
- b. It remains nearly constant in magnitude and position.
- c. It is easily calculated from the unit weight of materials.

$$\text{Dead Load} = \text{Unit Weight} \times \text{Volume}$$

Live load is the movable or variable load imposed on a structure during its use.

Examples:

- Occupants
- Furniture
- Vehicles on bridges
- Stored materials
- Movable machinery

Characteristics:

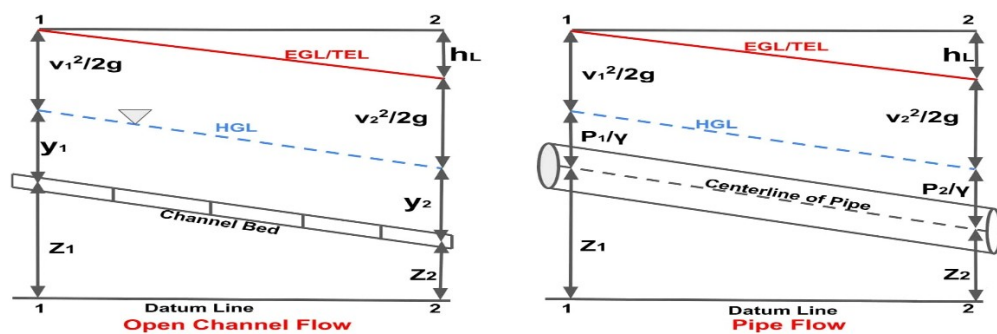
- Live load changes in magnitude and location with time.
- It depends on the type of occupancy of the building.
- The magnitude of live load is specified by building codes (e.g., IS 875 Part 2, NBC).

Thus, the structural loads are classified based on their nature and behavior, and proper consideration of these loads is essential for safe and economical design.

4. Differentiate between pipe and open channel flow. [5]

Answer:

Fluid flow can occur either through the closed conduits under pressure or through open channels with a free surface. There are two types of flow differing in their governing principles and behavior. They are listed below:



Pipe flow	Open channel flow
Flow of liquid in a closed conduit running in a full flow is known as pipe flow.	Flow of liquid with a free surface exposed to atmosphere is known as open channel flow.
Pipe flow occurs due to the pressure difference between two points	Open channel flow is due to the channel slope, where water flows under gravity.
The flow area in the pipe flow is fixed (pipe cross-section remains constant)	Flow area in open channel flow is variable (depends on depth of flow)
Pipe flow depends on pressure gradient, pipe roughness, pipe material and its diameter.	Open channel flow depends on slope, depth, channel roughness and hydraulic radius.
Pressure energy is significant in pipe flow.	Potential energy (elevation head) dominates in open channel flow.
Pipe flow is determined by Darcy-Weisbach, Hazen-Williams equations	Open channel flow is determined by Manning's equation, Chezy's equation etc.
The flow in the pipe always runs full.	The flow in open channel may be uniform or non-uniform.
The parameters that control the pipe flow are: pressure, diameter, and length of pipe	The parameters that control the open channel flow area: Depth, slope, and cross-sectional shape

Pipe flow	Open channel flow
Examples of pipe flow: Water supply pipelines, sewer pipes (running full)	Examples of open channel flow: Rivers, canals, drainage channels

Thus, pipe flow is pressure-driven and occurs in closed conduits, whereas open channel flow is gravity-driven with a free surface, leading to fundamentally different analysis and design approaches.

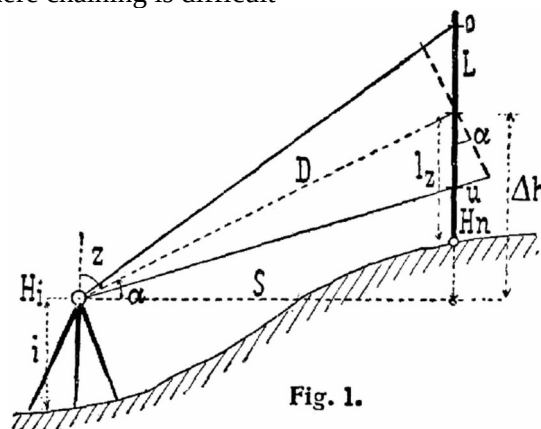
5. Write about the tacheometry. Explain the errors in stadia surveying and list the various methods of setting out a simple circular curve. [3+2+5=10]

Answer:

Tacheometry is the branch of surveying in which distances and elevations are obtained indirectly using a tacheometer based on optical principles. It is a rapid surveying method used to determine horizontal distances and elevations without direct measurement on the ground.

This method of surveying is used in rough and hilly terrains.

- Suitable for topographic surveying
- Helpful where chaining is difficult



The basic distance formula is:

$$D = Ks + C$$

where :

D = Horizontal distance

s = Staff intercept (difference between upper $\wedge$ lower stadia readings)

$$K = \frac{f}{i} = \text{Multiplying (stadia) constant (generally 100)}$$

$$C = f + d = \text{Additive constant (generally 0 for anallactic lens)}$$

For a horizontal line of sight, $D = 100s$ when $K = 100 \wedge C = 0$.

Stadia surveying is subjected to various errors that affect the accuracy of measured distances and elevations. These errors can be **instrumental, personal, and natural errors**.

- Instrumental errors:** Incorrect value of the stadia constant (K) or additive constant (C), improper adjustment of the stadia hairs, line of sight not truly horizontal when required.

- ii. **Personal errors:** Incorrect reading of the stadia staff, staff not held perfectly vertical, mistakes in booking or recording observations, improper focusing, leading to parallax.
- iii. **Natural errors:** Poor visibility caused by fog, rain, dust, or haze, atmospheric refraction affecting long sights, heat waves or shimmering air causing inaccurate readings, strong wind causing the staff to sway.

The diagram shows a diamond-shaped structure with vertices V (top), O (bottom), T_1 (left), and T_2 (right). A vertical dashed line passes through V and O . A horizontal line segment connects T_1 and T_2 . A point C is on the vertical line above the horizontal segment, and a point E is on the line VT_1 . A curved line segment connects E and T_2 . A horizontal line segment EE' is drawn from E to the vertical line, where E' is on the curve. A vertical double-headed arrow O_o is shown between E' and the horizontal line. A point F is on the horizontal line to the left of D , and a horizontal double-headed arrow x is shown between F and D . The distance from T_1 to F is labeled $L/2$, and the distance from D to T_2 is also labeled $L/2$. A point O_x is shown near E . Two lines labeled R connect O to T_1 and O to T_2 .

- Two theodolites used simultaneously
- Provides higher accuracy

5. **Coordinate method:**

- Points fixed using calculated coordinates
- Used in modern surveying

Thus, tacheometry provides rapid surveying, stadia surveying is affected by various errors, and several methods are available for accurately setting out a simple circular curve depending on site conditions.

6. **Describe in brief 'Darcy's law for laminar flow through saturated soil. [10]**

Answer:

Darcy's law was proposed by the French engineer **Henry Darcy** in **1856** based on experiments on water flow through sand filters.

Darcy's Law states that the rate of flow of water through a saturated soil is directly proportional to the hydraulic gradient and the cross-sectional area of the soil, provided the flow is laminar. It is expressed mathematically as:

$$Q = k i A$$

where :

$$Q = \text{Discharge } (m^3/s)$$

$$k = \text{Coefficient of permeability } (m/s)$$

$$i = \text{Hydraulic gradient}$$

$$A = \text{Cross-sectional area normal to the flow } (m^2)$$

$$\text{The seepage velocity is: } v = \frac{Q}{A} = ki$$

$$\text{where: } v = \text{Darcy's (discharge) velocity } (m/s)$$

Darcy's law is applicable only when:

- The soil is fully saturated.
- The flow is laminar.
- The soil is homogeneous and isotropic.
- The hydraulic gradient is constant.
- Water is incompressible and the temperature remains nearly constant.

Limitations of Darcy's law:

- Not valid for turbulent flow (common in coarse gravels).
- Not applicable to unsaturated soils.
- Deviations may occur in highly plastic clays at very low hydraulic gradients.

Applications:

- Seepage analysis in earth dams and embankments.
- Design of drainage systems and filters.

- c. Estimation of groundwater flow.
- d. Calculation of permeability of soils.
- e. Analysis of seepage beneath hydraulic structures.

Thus, Darcy's law provides the fundamental relationship between discharge and hydraulic gradient which is best in analyzing laminar flow through saturated soil in engineering practice.

Section-B (40 Marks)

7. How are labor, material and overhead costs considered into rate analysis according to the procurement Regulation, 2064? [5]

Answer:

According to the Public Procurement Regulation, labor, material, and overhead costs are integrated into the rate analysis for construction works using a specific standardized formula.

a. Base Rate Components:

The base rate of a specific work item is first determined by summing three primary costs:

- i. **Labor cost:**
Based on approved wage rates set by the Government of Nepal.
Includes wages of skilled, semi-skilled, and unskilled labor.
Productivity of labor (output per day) is considered.
Includes allowances, insurance, and statutory benefits if applicable.
- ii. **Cost of construction materials:**
Based on market rate or standard district rates.
Includes:
 - a. Basic cost of materials
 - b. Transportation (lead and lift)
- iii. **Machine and equipment fare:**
Includes hire charges of machinery and tools.
Covers fuel, maintenance, operator cost, and depreciation.

The standard rates for these components (labor wages, material prices, and equipment rent) are fixed annually in each district by the Rate Fixation Committee based on the prevailing market rate.

b. Consideration of Overhead:

Overhead is provided to cover the indirect project expenses like site supervision, administrative salaries, insurance, and utilities. Once the base rate is established, overhead is added as follows:

- Bidder's overhead is calculated as 15 percent of the combined base rate i.e., 10% for contractor's overhead and 5% for profit.
- Total rate for an item is reached by multiplying the base rate by 1.15 (Base Rate + 15% Overhead).

Method of finding out the rate of item:

Labor cost (a)

Cost of construction materials (b)

Machine, equipment fare (including oil and lubricants) (c) 253

Rate of item (a + b+ c) = (d)

Bidder's overhead (15 percent of d) =0.15 (d)

Total rate (e) = 1.15 (d)

c. Final Cost Estimate Calculation

After determining the total rate for all items, additional percentage-based costs are considered to find the total cost estimate:

- **Work chart staff costs:** Added at a rate of 2 percent of the total item costs.
- **Other small/petty costs:** Added at a rate of 2.5 percent of the total item costs.
- **Contingencies:** For budget management, the total project cost may also include a 10 percent physical contingency and a 10 percent price adjustment contingency.
- **Value Added Tax (VAT):** The final total cost estimate for a project or program must also include the applicable VAT.

In the case of consultancy services, a different format is used where costs are categorized into human resource remuneration, reimbursable expenses (travel, equipment), and miscellaneous expenses (training, printing), with an added office management expense typically set at five percent.

Section-C (25 Marks)

8. Why estimate is essential in construction work? State the types of estimates. [5]

Answer:

An estimate is the calculation of the quantities of materials, labor, equipment, and the probable cost of a construction project before the work begins. It is an essential part of project planning and management.

Estimate is essential in construction work for the following reasons:

- i. **Determines the Total Cost:** Estimates the probable cost of the project before execution.
- ii. **Tendering and Contracting:** Estimate forms the basis for inviting tenders, comparing bids, and awarding contracts.
- iii. **Quantity of Materials:** Estimate determines the quantities of materials, labor, and equipment required for the project.
- iv. **Project Planning and Scheduling:** It assists in planning the sequence of work, resource allocation, and construction schedule.
- v. **Cost Control:** Estimate enables comparison between estimated and actual costs to control project expenses.

Types of estimates:

- i. **Preliminary (Approximate or Rough Cost) Estimate:**
 - Prepared at the initial stage to determine the approximate cost of a project.
 - Used for feasibility studies, budgeting, and administrative approval.
- ii. **Detailed Estimate:**
 - Prepared after the final drawings and specifications are completed.
 - Includes detailed quantity calculations and item-wise cost.
 - Used for technical sanction, tendering, and execution.
- iii. **Revised Estimate:**
 - Prepared when the cost of the original estimate is likely to exceed the sanctioned amount due to changes in quantities, rates, or design.
 - Submitted for revised approval.
- iv. **Supplementary Estimate**
 - Prepared for additional works not included in the original estimate but found necessary during construction.
 - It is prepared in addition to the original estimate.

9. Neatly sketch the following types of RCC footings and explain the situation in which they are used. a) Pile foundation b) Mat foundations. [1+3+3+3]

Answer:

RCC (Reinforced Cement Concrete) footings are structural elements that transfer loads from columns or walls to the ground. The selection of footing depends on the soil bearing capacity, column arrangement, and structural loading.

a. Pile foundation:

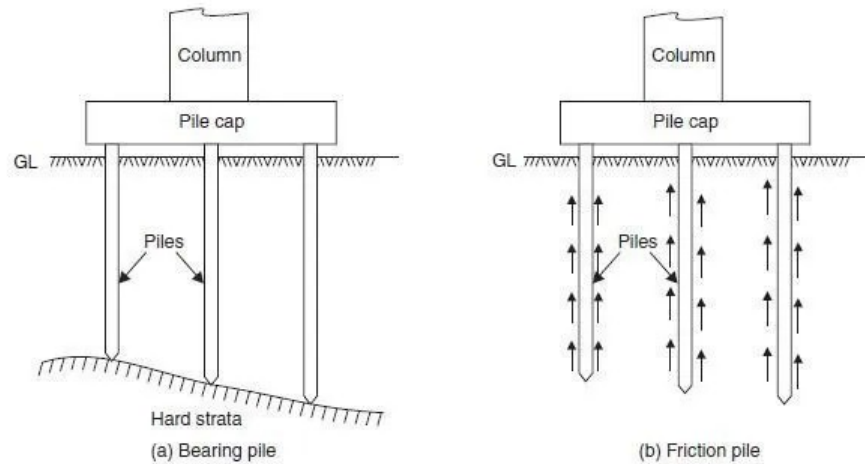
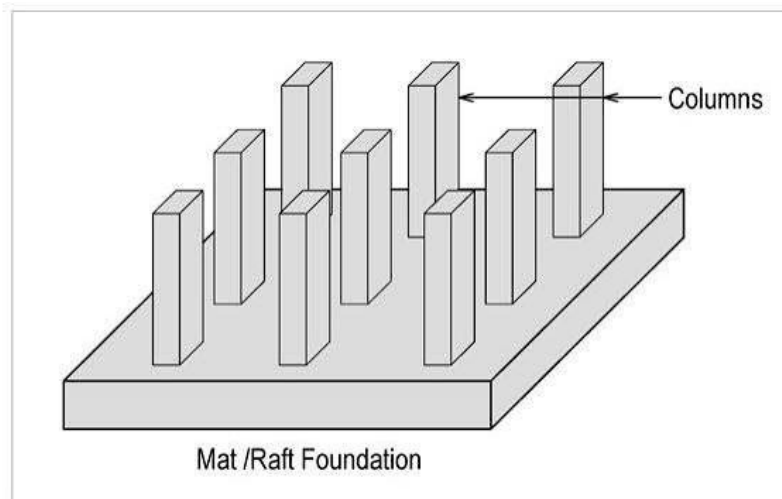


Fig. 7.9. Pile foundations

- Used when surface soil is weak or compressible.
- Transfers load to deeper strong strata
- Suitable for:
 - a. Waterlogged areas
 - b. Heavy structures (bridges, high-rise buildings)
 - c. Also resists uplift and lateral loads

b. Mat foundations:



- Used when soil bearing capacity is low.
- Suitable when columns are closely spaced.
- Spreads load over large area.
- Reduces differential settlement.
- Economical for multi-storey buildings on weak soil.
- Mat foundations are used where the foundation covers more than about 50% of the building area, making a raft foundation more economical than individual footings.

10. List out the various types of failures in flexible pavement. Write down the operational procedures in sequential order for patch repair of potholes. [6+4=10]

Answer:

Flexible pavement failures are defects or distresses that occur due to repeated traffic loading, poor drainage, inadequate design, poor construction, or environmental effects.

a. Alligator cracking:

common type of flexible pavement failure that occurs due to the relative movement of pavement layer materials.

b. Formation of waves and corrugations:

occurs due to poor subgrade conditions, poor gradation of mix, unstable underlying layers.

c. Shear failure and cracking:

occurs due to inherent weakness of pavement mix, excessive loading.

d. Bleeding: Excess bitumen appears on the surface, making it smooth and slippery.

e. Shoving: Localized displacement of pavement caused by braking or acceleration.

f. Longitudinal Cracking: Cracks parallel to the pavement centerline due to poor construction joints or shrinkage.

g. Transverse Cracking: Cracks perpendicular to the centerline caused by temperature changes or shrinkage.

h. Stripping: Separation of bitumen from aggregates due to moisture.

Patch repair is a maintenance technique used to restore the structural integrity and riding quality of a pavement by filling potholes with suitable bituminous material.

Sequential orders for patch repair of potholes:

1. Inspect and Mark the Pothole

- Identify the damaged area and mark the repair boundary.

2. Cut the Pothole to a Regular Shape

- Trim the edges into a rectangular or square shape with vertical sides using a pavement cutter or chisel.
- Remove all loose and deteriorated material.

3. Clean the Pothole

- Remove dust, loose particles, and standing water using a broom, compressed air, or wire brush.
- Ensure the surface is clean and dry.

4. Apply Tack Coat

- Apply a thin coat of bitumen emulsion or hot bitumen on the sides and bottom of the pothole to ensure proper bonding between the old pavement and the new patch.

5. Place the Patching Material

- Fill the pothole with suitable hot mix asphalt (preferred) or cold mix bituminous material.
- Place the material in layers if the pothole is deep.

6. Compact the Patch

- Compact each layer using a hand rammer, plate compactor, or roller until the patch is dense and stable.
- Slightly overfill before compaction so that the finished patch is level with the surrounding pavement.

7. Finish the Surface

- Ensure the repaired surface is flush with the existing pavement and provides a smooth riding surface.
- Seal joints if necessary to prevent water infiltration.

8. Open to Traffic

- Allow the patch to cool and stabilize (for hot mix) before opening the road to traffic.

Section-D (25 Marks)

11. What are the objectives of water treatment? Simply list out the various stages of water treatment system in chronological order with type of impurity treated in each stage. Explain 'chlorination' method of disinfection in brief. [3+4+3=10]

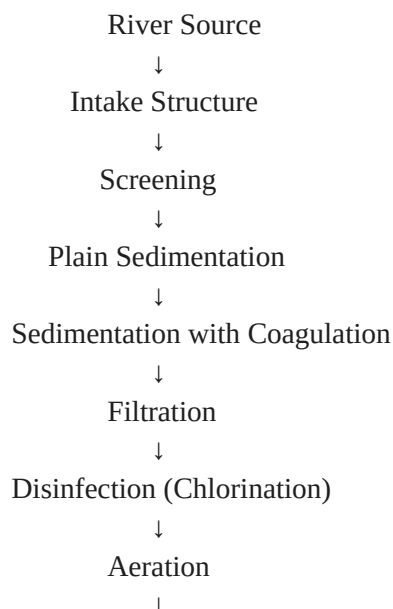
Answer:

Water treatment is the process of removing physical, chemical, and biological impurities from raw water to make it safe and suitable for drinking and other domestic uses.

Objectives of water treatment:

- Remove Suspended and Floating Matter**
 - Water treatment helps to eliminate sand, silt, clay, algae, and other suspended impurities.
- Remove Disease-Causing Microorganisms**
 - It also helps in destroying bacteria, viruses, protozoa, and other pathogens to prevent water-borne diseases.
- Improve Taste, Odor, and Color**
 - It also removes objectionable color, unpleasant taste, and foul odor to make water aesthetically acceptable.
- Reduce Turbidity**
 - It produces clear and transparent water by removing fine suspended particles.
- Provide Safe and Potable Water**
 - It ensures the treated water meets national and international drinking water quality standards (e.g., WHO guidelines).

Schematic flow diagram illustrating the possible treatment units involved in the water treatment process:



Softening



Miscellaneous

Screening:

- Screening is the process of passing water through screens to remove large floating and suspended matter such as sticks, branches of tree, dead animals etc.

Plain sedimentation:

- The process of removal of coarse suspended particles present in the water such as sand, silt by allowing it to remain quiescent in sedimentation tank so that the suspended material settle down at the bottom by the action of gravity is known as plain sedimentation.

Sedimentation with coagulation:

- The fine suspended and colloidal materials like clay particles which cannot be removed by plain sedimentation or takes long time to removal by plain sedimentation can be removed by sedimentation with coagulation.
- Coagulants such as aluminum sulphate, sodium aluminate, iron salts are used.

Filtration:

- Filtration is the process of passing water through the beds of sand and granular material.
- For the removal of those impurities whose specific gravity is less than or equal to that of water for which sedimentation process does not work, in such case filtration is used.

Disinfection:

- The process of killing pathogenic bacteria and microorganisms present in water for the safe use is known as disinfection.
- Chlorination is one of the most common processes of disinfecting water.

Aeration:

- The process of bringing water in contact with air to remove dissolved gasses present in water is known as aeration.
- Aeration removes CO₂ gasses up to 70% due to which corrosion corrosiveness of water is reduced.

Softening:

- The process of removal or reduction of temporary or permanent hardness of water is known as softening.
- Softening helps to improve the taste of water, reduce consumption of soap during washing, bathing and also reduces formation of scales in boiler.

Miscellaneous:

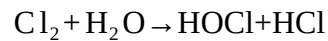
- It includes:
 - Removal of iron and manganese
 - Removal of color, odor, taste from the water
 - Removal of arsenic and so on.

Chlorination:

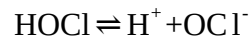
Chlorination is the most common and economical method of disinfecting drinking water. It involves adding **chlorine or chlorine compounds** to water to destroy disease-causing microorganisms such as bacteria, viruses, and some protozoa, making the water safe for human consumption.

Principle:

When chlorine is added to water, it reacts to form **hypochlorous acid (HOCl)** and **hydrochloric acid (HCl)**:



Hypochlorous acid further dissociates:



The hypochlorous acid (HOCl) is the main disinfecting agent. It penetrates microorganisms and destroys their enzymes and cell walls, thereby killing them.

Types of chlorination:

- i. **Plain chlorination:** Chlorine is added to relatively clear water.
- ii. **Pre-chlorination:** Chlorine is added before sedimentation and filtration.
- iii. **Post-chlorination:** Chlorine is added after filtration (most common).
- iv. **Break-point chlorination:** Chlorine is added until all impurities are oxidized and a free residual chlorine remains.
- v. **Super-chlorination:** A high dose of chlorine is applied, followed by de-chlorination if required.

12. Express in brief the different types of irrigation canal alignment with merits and demerits of each. [10]

Answer:

Canal alignment is the route or course selected for an irrigation canal to convey water from the source to the command area economically and efficiently.

Types of irrigation canal alignment are:

- i. Ridge Canal Alignment
- ii. Contour Canal Alignment
- iii. Side Slope Canal Alignment
- iv. Valley canal Alignment

i. Ridge canal alignment:

A canal aligned along the ridge (watershed) separating two drainage basins.

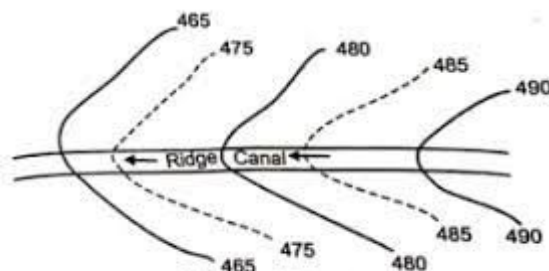


Fig. 6.3 : Ridge Canal

Advantages:

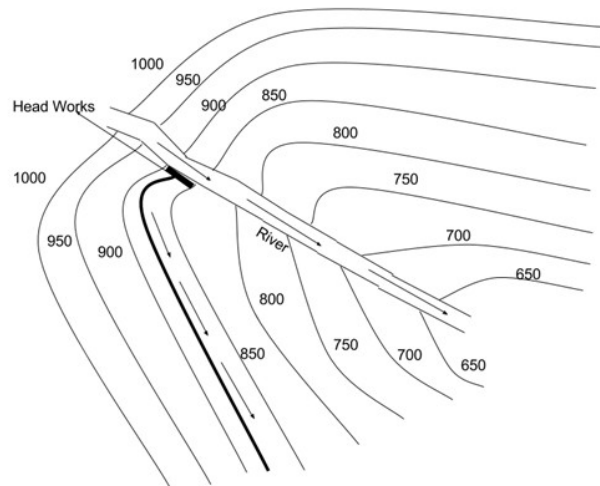
- a. Irrigates land on both sides of the canal.
- b. Requires fewer cross-drainage works.
- c. Less chance of waterlogging.
- d. Gravity irrigation is possible over a large area.

Disadvantages:

- a. May require heavy cutting in hilly terrain.
- b. Longer canal length in some cases.
- c. Initial construction cost may be high.

ii. Contour Canal Alignment:

A canal constructed approximately along a contour line, maintaining nearly the same ground elevation.



Advantages:

- a. Minimum excavation and embankment.
- b. Suitable for hilly and undulating terrain.
- c. Lower construction cost in such areas.

Disadvantages:

- a. Longer canal length due to winding alignment.
- b. Requires more cross-drainage structures.
- c. Difficult operation and maintenance.

iv. Side slope canal alignment:

A canal laid along the side slopes of hills at a suitable gradient.

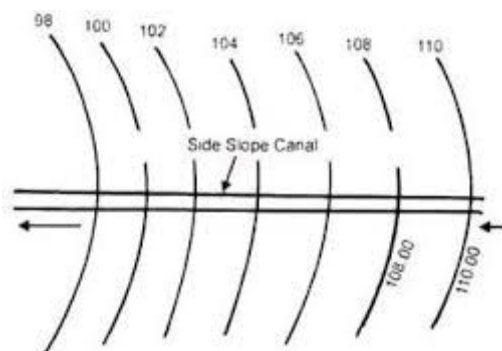


Fig. 16.3. Side slope channel

Advantages:

- a. Suitable for mountainous and steep regions.
- b. Gravity flow can be maintained.
- c. Serves hillside agricultural lands effectively.

Disadvantages:

- a. Risk of landslides and slope failures.
- b. Costly construction and maintenance.
- c. Requires retaining walls and protective works.

v. **Valley Canal Alignment:**

A canal aligned along the valley floor or near the lowest part of a valley.

Advantages:

- a. Shorter canal length.
- b. Easy construction due to gentle terrain.
- c. Less excavation in some cases.

Disadvantages:

- a. Requires numerous cross-drainage structures.
- b. Difficult to irrigate higher lands by gravity.
- c. Greater risk of flooding and waterlogging.

13. कोशी प्रदेश सरकार अन्तर्गत प्रत्येक मन्त्रालय वा निकायमा रहने लेखापरीक्षण तथा आन्तरिक नियन्त्रण समितिको गठन के कसरी हुने व्यवस्था रहेको छ ? लेख्नुहोस् । [5]

Answer:

प्रदेश आर्थिक कार्यविधि तथा वित्तीय उत्तरदायित्व ऐन, 2078 को दफा 34 बमोजिम कोशी प्रदेश सरकार अन्तर्गतका प्रत्येक मन्त्रालय वा निकायमा आन्तरिक नियन्त्रण प्रणाली, आन्तरिक तथा अन्तिम लेखापरीक्षण प्रतिवेदनको कार्यान्वयनका लागि देहाय बमोजिमको "लेखापरीक्षण तथा आन्तरिक नियन्त्रण समिति" रहने व्यवस्था गरिएको छ :

- **संयोजक:** सम्बन्धित मन्त्रालय वा निकायको लेखा उत्तरदायी अधिकृत (प्रदेश सचिव) वा निजले तोकेको अधिकृत ।
- **सदस्य:** सम्बन्धित निकायको योजना तथा अनुगमन महाशाखा प्रमुख ।
- **सदस्य:** सम्बन्धित निकायको प्रशासन महाशाखा प्रमुख ।
- **सदस्य:** संयोजकले तोकेको सम्बन्धित निकायमा कार्यरत विषय क्षेत्रको ज्ञान तथा अनुभव भएको सम्बन्धित पदाधिकारी।
- **सदस्य-सचिव:** सम्बन्धित निकायको आर्थिक प्रशासन प्रमुख ।

समितिको प्रमुख कार्यहरू समितिका मुख्य कार्यहरू निम्न छन्:

- आन्तरिक नियन्त्रण प्रणाली स्थापना र सुदृढिकरण गर्ने
- लेखा परीक्षण (Internal Audit) को प्रतिवेदन अध्ययन गर्ने
- त्रुटि, कमजोरी तथा अनियमिता पहिचान गर्ने
- सुधारका लागि आवश्यक सिफारिस गर्ने
- बेरुजु फछ्यौट तथा अनुगमन गर्ने
- वित्तीय अनुशासन कायम गर्ने
- नीति, नियम तथा कार्यविधिको पालना सुनिश्चित गर्ने

यस समितिले मुख्यतया मन्त्रालय वा निकाय र मातहतका कार्यालयहरूबाट सम्पादन गरिने कार्यहरू दक्षतापूर्ण र मितव्ययी रूपमा सम्पन्न गराउने, वित्तीय जोखिम न्यूनीकरण गर्ने, वित्तीय विवरणहरूलाई विश्वसनीय बनाउने र लेखापरीक्षणबाट कायम भएका बेरुजुहरू फछ्यौट गरी आन्तरिक नियन्त्रण प्रणालीलाई सुदृढ एवं प्रभावकारी बनाउन आवश्यक कार्यहरू गर्दछ।

14. प्रदेश सुशासन (व्यवस्थापन तथा सञ्चालन) ऐन, 2076 (कोशी प्रदेश) अनुसार कार्यालय प्रमुखलाई के कस्ता काम, कर्तव्य र अधिकार तोकिएका छन् ? उल्लेख गर्नुहोस् । [5]

Answer:

प्रदेश सुशासन (व्यवस्थापन तथा सञ्चालन) ऐन, 2076 (कोशी प्रदेश) को दफा 12 मा कार्यालय प्रमुखको जिम्मेवारी र कर्तव्यसम्बन्धी स्पष्ट व्यवस्था गरिएको छ।

कार्यालय प्रमुखको काम, कर्तव्य र अधिकार यस ऐनअनुसार आफ्नो कार्यालयको प्रशासनिक प्रमुखको रूपमा कार्य सम्पादन गर्दा कार्यालय प्रमुखको मुख्य जिम्मेवारी प्रशासनलाई छिटो, छरितो, मितव्ययी र नतिजामुखी बनाउनु हुनेछ।

यसका विस्तृत कार्यहरू निम्नानुसार छन्:

- **प्रशासनिक व्यवस्थापन:**

कार्यालयको दैनिक प्रशासन सञ्चालन गर्ने र कार्यालयमा अनुशासन, ढिलासुस्ती नियन्त्रण एवं कार्यक्षमता कायम गर्ने।

➤ **बजेट तथा कार्यक्रम कार्यान्वयन:**

स्वीकृत वार्षिक कार्यक्रम र बजेटको अधीनमा रही तोकिएको समय सीमाभित्र निर्धारित लक्ष्य हासिल गर्ने गरी कार्य सम्पादन गर्ने।

➤ **कर्मचारी परिचालन:**

मातहतका कर्मचारीहरूको कामको बाँडफाँड गर्ने, उनीहरूको कामको सुपरिवेक्षण गर्ने र आवश्यक निर्देशन दिने।

➤ **जिम्मेवारी र जवाफदेहिता:**

आफू र मातहतका कर्मचारीलाई तोकिएको कामप्रति जिम्मेवार र जवाफदेही बनाउने।

➤ **सेवा प्रवाह र गुणस्तर:**

कार्यालयाबाट प्रदान गरिने सेवा वा वस्तुको गुणस्तर सुनिश्चित गर्ने र सेवाग्राहीले सुलभ तरिकाले सेवा पाउने वातावरण निर्माण गर्ने।

➤ **नागरिक बडापत्रको पालना:**

कार्यालयमा नागरिक बडापत्र (Citizen Charter) राख्ने र त्यसमा उल्लेख भए बमोजिमको सेवा तोकिएको समय र प्रक्रियाभित्र उपलब्ध गराउने।

➤ **गुनासो व्यवस्थापन:**

सेवाग्राहीका गुनासोहरू सुन्ने र त्यसको तत्काल सम्वन्ध गर्ने संयन्त्रको विकास गर्ने।

➤ **अभिलेख र भौतिक सम्पत्तिको संरक्षण:**

कार्यालयको सरकारी चल-अचल सम्पत्ति, कागजात र अभिलेखहरूको उचित संरक्षण र व्यवस्थापन गर्ने।

➤ **अनुगमन र प्रतिवेदन:**

कार्यालयको कार्य प्रगतिको नियमित अनुगमन गर्ने र तोकिएबमोजिम आवधिक प्रगति प्रतिवेदन माथिल्लो निकाय वा अधिकारी समक्ष पेश गर्ने।

➤ **अधिकार प्रत्यायोजन:**

आफूमा निहित अधिकारमध्ये आवश्यक देखिएमा केही अधिकार मातहतका कर्मचारीलाई प्रत्यायोजन गर्ने।

15. स्थानीय सरकारी सेवामा कार्यरत कर्मचारीले कति किसिमका विदा पाउन सक्छन् ? उल्लेख गर्नुहोस् । [5]

Answer:

स्थानीय सरकार संचालन ऐन, 2074 को दफा 86 ले स्थानीय सेवाको गठन र सञ्चालन सम्बन्धी व्यवस्था गरेको छ। यसै अन्तर्गत बनेका स्थानीय सेवा (गठन तथा सञ्चालन) नियमावलीहरू र प्रचलित निजामती सेवाका मापदण्डहरू अनुसार स्थानीय सरकारी सेवामा कार्यरत कर्मचारीले पाउने विदाका प्रकार र अवधिलाई निम्नानुसार उल्लेख गर्न सकिन्छ :

○ **भैपरी आउने र पर्व विदा (Casual and festival leave):**

कर्मचारीले प्रत्येक वर्ष भैपरी आउने विदा 6 दिन र पर्व विदा 6 दिन गरी जम्मा 12 दिन विदा पाउनेछन्।

○ **घर विदा (Home leave):**

कर्मचारीले काम गरेको अवधिको 12 दिनमा 1 दिनका दरले वर्षको 30 दिन घर विदा पाउनेछन्।

○ **बिरामी विदा (Sick leave):**

प्रत्येक कर्मचारीले प्रत्येक वर्ष 12 दिन बिरामी विदा पाउनेछन्। यो विदा सञ्चित गरी राख्न सकिन्छ र 7 दिन भन्दा बढी अवधिको विदा माग गर्दा चिकित्सिकाको प्रमाणपत्र पेश गर्नु पर्दछ।

○ **प्रसूति विदा (Maternity leave):**

महिला कर्मचारी गभविती भएमा सुत्केरीको अघि र पछि गरी जम्मा 98 दिन प्रसूति विदा पाउनेछन्। यो विदा सेवा अवधिभरमा दुई पटक मात्र पाइनेछ।

- **प्रसूति स्याहार विदा (Paternity leave):**
कुनै पुरुष कर्मचारीको पत्नी सुत्केरी हुने भएमा त्यस्तो कर्मचारीले 15 दिन प्रसूति स्याहार विदा पाउनेछन् ।
- **किरीया विदा (Obsequies leave):**
कर्मचारी आफै किरीया बस्नु परेमा वा कुल धर्म अनुसार किरीया बस्नु पर्ने भएमा 15 दिन किरीया विदा पाउनेछन् ।
- **अध्ययन विदा (Study leave):**
स्थानीय सेवामा कम्तीमा 3 वर्ष सेवा अवधि पूरा गरेका कर्मचारीले सेवा अवधिभरमा एकै पटक वा पटक-पटक गरी 3 वर्ष र आवश्यकता अनुसार थप 2 वर्ष गरी जम्मा 5 वर्षसम्म अध्ययन विदा पाउन सक्छन् ।
- **असाधारण विदा (Extraordinary leave):**
कुनै कर्मचारीलाई विशेष कारणले विदा आवश्यक परेमा एक पटकमा 1 वर्ष नबढाई सेवा अवधिभरमा बढीमा 3 वर्षसम्म असाधारण विदा दिन सकिने व्यवस्था छ।
- **सट्टा विदा (Substitute leave):**
सार्वजनिक विदाका दिनमा काममा खटिएका कर्मचारीले त्यसको बदलामा पाउने विदालाई सट्टा विदा भनिन्छ। यस्तो विदा 1 वर्षभित्र लिइसक्नुपर्ने व्यवस्था छ।
- **सार्वजनिक विदा (Public holidays):**
नेपाल सरकार, प्रदेश सरकार वा सम्बन्धित स्थानीय तहले तोकेका सार्वजनिक विदाहरूमा कर्मचारीले विदा पाउनेछन् ।

निष्कर्षमा, विदा कर्मचारीको अधिकार नभई एक सुविधा मात्र हो। माथि उल्लेखित विदाका प्रकारहरूले स्थानीय तहमा कार्यरत कर्मचारीहरूको व्यक्तिगत, सामाजिक र वृत्ति विकासका आवश्यकताहरूलाई सम्बोधन गर्न सहयोग पुर्याउँछन् ।

16. नेपालको संविधानले प्रदेश सभाको विशेषाधिकार अन्तर्गत के कस्ता अधिकारको व्यवस्था गरेको छ ? स्पष्ट पार्नुहोस् ।

Answer:

नेपालको संविधानको धारा 187 मा प्रदेश सभाको विशेषाधिकार सम्बन्धी व्यवस्था गरिएको छ। प्रदेश सभालाई आफ्नो कार्य सम्पादनमा स्वतन्त्रता र निर्भयता प्रदान गर्न निम्न अनुसारका विशेषाधिकारहरूको व्यवस्था गरेको छ:

- **पूर्ण वाक स्वतन्त्रता:**
प्रदेश सभामा पूर्ण वाक स्वतन्त्रता रहनेछ। सभामा व्यक्त गरेको कुनै कुरा वा दिएको कुनै मतको सम्बन्धमा कुनै पनि सदस्यलाई अदालतमा कारबाही चलाइने छैन।
- **अदालतमा कारबाहीबाट संरक्षण:**
प्रदेश सभाको अधिकारले प्रकाशित गरेको कुनै प्रतिवेदन, पत्र, मत वा कार्यविधि (Proceedings) को सम्बन्धमा कुनै व्यक्ति उपर कुनै अदालतमा कारबाही चलाइने छैन।
- **आन्तरिक विषयको निर्णय गर्ने अधिकार:**
प्रदेश सभालाई आफ्नो आन्तरिक कामकारबाही नियमित गर्ने पूर्ण अधिकार हुनेछ । सभाको कुनै काम कारबाही नियमित छ वा छैन भनी निर्णय गर्ने अधिकार सम्बन्धित सभालाई मात्र हुनेछ र यस विषयमा कुनै अदालतमा प्रश्न उठाइने छैन ।
- **अदालतको हस्तक्षेप नहुने:**
प्रदेश सभाको कुनै कामकारबाहीको नियमिततामाथि छानबिन गर्ने वा सोधपूछ गर्ने अधिकार कुनै पनि अदालतलाई हुने छैन।
- **अवहेलनामा कारबाही गर्ने सक्ने:**
प्रदेश सभाको विशेषाधिकारको हनन्लाई सभाको अवहेलना मानिनेछ। कुनै सदस्य वा अन्य व्यक्तिले सभाको अवहेलना गरेमा सभामुखको निर्णयले त्यस्तो व्यक्तिलाई सचेत गराउने, नसिहत दिने वा निश्चित अवधिसम्म कैद वा जरिवाना गर्न सक्ने अधिकार सभालाई हुन्छ।

➤ **प्रदेश कानून बमोजिम हुने:**

विशेषाधिकारका अन्य विषय प्रदेश कानूनले निर्धारण गरे बमोजिम हुनेछन् ।यी व्यवस्थाहरूले प्रदेश सभालाई जनउत्तरदायी भई स्वतन्त्र रूपमा कानून निर्माण र शासन प्रणालीको निगरानी गर्न सक्षम बनाउँछन् ।